

Unit – 3 Introduction of Enterprise Resource Planning (ERP)

Concept of EMS and ERP

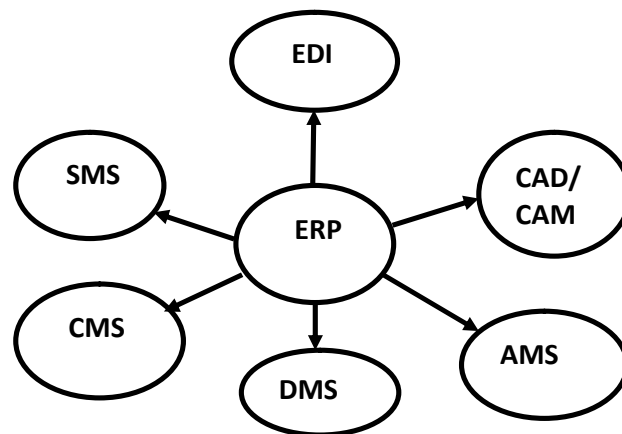
EMS (Enterprise Management Systems) Introduction:-

- Due to emerging trends in information technology and audio-video technology has changed the style of business operations.
- The word enterprise expresses the large business community with its business associates. Hence the management of information system should consider the wide aspect of enterprise. Any information related to business must be communicated immediately across the enterprise for update like online system.
- The operations of all departments/ centers are to be integrated, coordinated and fully transparent for taking suitable actions and decisions. The information system must be automated and interfaced to other systems by using wireless communications. Remote sensing system for security and communications may be used.

Enterprise Management Systems (EMS):-

Various means, tools, techniques systems and technologies are available to meet the business needs. In order to support the enterprise functioning, there is need of an integrated solution of these means (tools, techniques, systems & technological) such integrated solution is called as Enterprise Management System(EMS). In an EMS system ERP(Enterprise Resource Planning) plays an important role.

Components of EMS-(Enterprise Management System)



Different essential components of EMS are:

- 1 ERP (Enterprise Resource Planning)
- 2 Electronic Commerce (EDI- Electronic Data Exchange/Interchange)
- 3 Design and Engineering (CAD – Computer Aided Designing / CAM – Computer Aided Manufacturing)
- 4 Attendance Data Management System (AMS – Attendance Management System)
- 5 Document Management System (DMS)
- 6 Communication Management System (CMS)
- 7 Security Management System (SMS)

1. ERP- Enterprise Resource Planning:-

ERP System plays an important role in EMS(Enterprise Management System). All major decisions and its execution are done through ERP. It provides required inputs for planning and control, it handles operational systems even. ERP takes support from six other systems for managing overall system. Each support systems has unique & independent functions to perform. The support systems

are EDI, CAD/CAM, AMS, DMS, SMS & CMS. Each system processes, capture, store and transfer the data to ERP.

2. EDI- Electronic Data Exchange:-

Electronic Data Exchange (EDI) is an electronic exchange of business transaction documents amongst the trading partners. EDI is designed for commercial function of business i. e. electronic commerce. EDI is electronically connected to ERP by E-mail, data transfer and document transfer. EDI interconnects customers and suppliers and other associated partners of suppliers.

3. CAD/CAM- Computer Aided Design / Computer Aided Manufacturing:

These systems handle design engineering by providing design engineering information to ERP which is used for manufacturing, purchase and inspection function. They are also supported with suitable databases for ERP.

4. AMS- Attendance Management System:

AMS maintains information of employee need for personnel planning and scheduling. It provides dynamic information about employee i.e. presence, shift rotation, kind of job handled.

5. DMS- Document Management System:

Important documents are maintained by DMS. The database is used for record of transactions handled, sending messages. The system provides text edit facilities, cross checking, confirmation of authenticity of documentation. DMS also provide facilities of broadcasting (sending document to multiple destinations), scanning, imaging and copying facility.

6. CMS- Communication Management System:

ERP uses CMS as a tool for its communication needs. CMS control overall communication within the enterprise.

7. SMS- Security Management System:

It handles security aspects of the enterprise. It tracks and records the movement of man, material and vehicle. Access by unauthorized or unidentified means are monitored and warned, so that the entire business operations can be safeguarded. In-time and out-time of each employee and materials with vehicle are recorded and sent to ERP.

EMS Components and their functions are summarized in the following table

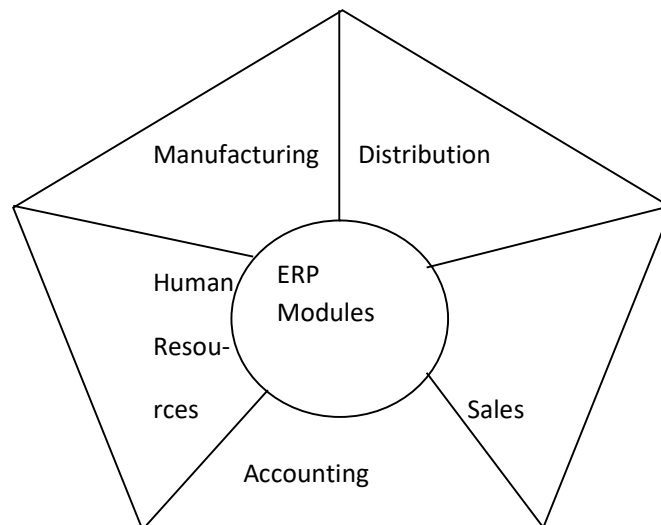
SR. No.	Component	Function
1	ERP	Managing all the functions of business with information support.
2	EDI	E- commerce, communication.
3	CAD/CAM	Design of Engineering, Production Management.
4	AMS	Employee Attendance.
5	DMS	Copying, Editing, Imaging and Broadcasting
6	CMS	Audio – Video Communication.
7	SMS	CCTV (Closed Circuit T.V.), alarming system.

ERP Systems

- ✓ ERP is a cross-functional enterprise system which deals with planning and use of resources in the business. The resources are material, manufacturing capacity manpower and capital. ERP helps to execute strategies, plan, decisions and actions within given timeframe, ERP supports core business processing, inventory management.
- ✓ ERP provide the enterprise an integrated real-time view of its core business process. It provides information and knowledge base to be used for planning and control of business operations. For this ERP is assisted by several support systems. The support systems may be a stand-alone system or from a part of a manufacturing system.

ERP Modules:

- ERP solutions are available in modules to suit the varying need fo business enterprise. The ERP solutions are available on UNIX and on Windows NT platforms. Typical application modules of an ERP package are:
 1. Manufacturing (Production & Planning)
 2. Distribution (Order Management, Purchasing, Logistics)
 3. Sales (Sales Analysis, Sales Planning, Pricing)
 4. Accounting (Book keeping, Managerial accounting)
 5. Human Resource (Manpower Planning, Salary Administration)



ERP MODULE

- These modules are designed for
 - ✓ Data capturing
 - ✓ Data transaction
 - ✓ Data validation
 - ✓ Analysis
 - ✓ Updating of data
 - ✓ Reporting
- ERP System tracks business resources through modules and the state of commitment made by business. Any module can enter the data into the system corresponding data in other module is automatically updated.
- Few important features of ERP are listed below. The features safeguard the business operation from illegal operation and protect the data from misuses.
 1. Security
 2. Authorization

3. Referencing
4. Implementation of business rules

ERP Architecture

- ❖ The ERP architecture is client / server and users object oriented technology for design and development of the system. There are three basic components of any information systems:
 - I) Data Management
 - II) Application Logic
 - III) Presentation
- ❖ Depending on the function client is a user and server provides the services to run the system. There exist three types of ERP architecture:
 - I) Two tier architecture
 - II) Three tier architecture
 - III) Mix architecture

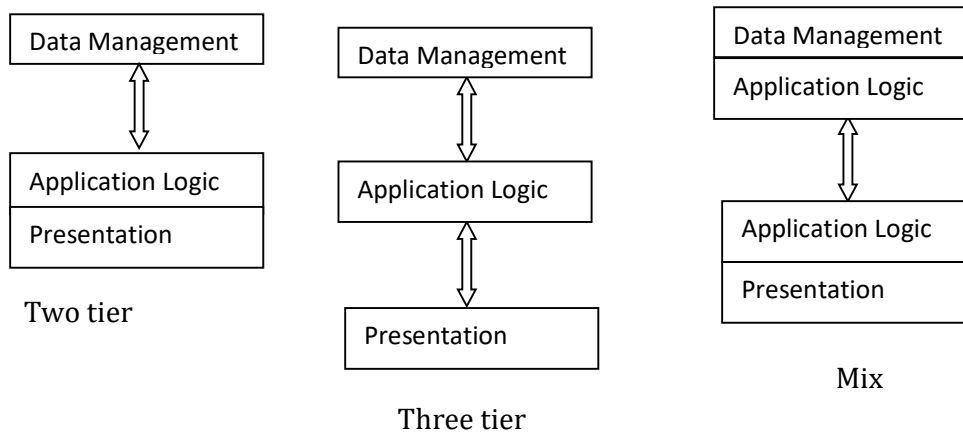


Figure: client / server model architecture

- ❖ As per the need of business operations suitable architecture is chosen. In two tier architecture all the requirements of users are sent to the server. Hence the load on the server is increased and response time is increased. When the user is dumb and uses the system in guided manner with help. Then three tier systems are useful. Mix architecture has partial advantage of both types. The best architecture is user friendly easy to understand and easy to maintain.

ERP Features

ERP feature can be categorized as per the module or function e.g. general function, business system, sales, production, material, finance, personal, fixed asset, maintenance, quality-control, consolidation of business operation. Each function has distinct features.

1. General Features:
 - a. Separation of code and data
 - b. Diagnostic functions
 - c. Help functions
 - d. Word processing / text editing
 - e. Enterprise Modeling
 - f. Integrated Solution for all requirements
2. Business Systems:
 - a. Business Forecasting
 - b. Target Setting
 - c. Business Planning

- d. Strategy Formulation
 - e. Monitor – control
 - f. Information – knowledge base
- 3. Sales:
 - a. Order Processing
 - b. Order analysis
 - c. Stores Management
 - d. Receivables
 - e. Market Research
 - f. Distributor Management
- 4. Production:
 - a. Production planning and control
 - b. Bill of material (costing)
 - c. Scheduling
 - d. Quality Standards
 - e. Progress Reports
 - f. Work station management
- 5. Materials:
 - a. Purchasing
 - b. Stock Management
 - c. Inventory
 - d. Quotations
 - e. Material Accounting
- 6. Finance:
 - a. Accounting
 - b. Ledger
 - c. Capital Management
 - d. Cost Accounting
 - e. Cost Analysis
- 7. Personnel:
 - a. Attendance record
 - b. Payroll account
 - c. Manpower planning
 - d. Training
- 8. Maintenance:
 - a. Maintenance planning and management
 - b. Contract management
 - c. Monitoring performance
 - d. Breakdown
- 9. Quality Control:
 - a. Data collection for quality
 - b. Quality assurance
 - c. Analysis of quality
 - d. Monitoring quality

Characteristics:

ERP (Enterprise Resource Planning) systems typically include the following characteristics:

- ❖ An integrated system that operates in real time (or next to real time), without relying on periodic updates.
- ❖ A common database, which supports all applications.
- ❖ A consistent look and feel throughout each module.

- ❖ Installation of the system without elaborate application/data integration by the Information Technology (IT) department.
- ❖ Packaged software
- ❖ Integrate the majority of a business's processes
- ❖ Process the majority of an organizations transactions
- ❖ Use a data warehouse
- ❖ Allow access to data in "real time"
- ❖ Integrate transaction processing and planning activities

Benefits of ERP

ERP system provides several benefits to the organization. The benefits of ERP can be classified as-

1. Tangible benefits:

The benefits which can be quantified or measured are called tangible benefits. Tangible benefits of ERP includes-

- I) Reduced lead- time
- II) Manifold increase in business
- III) Improved inventory system
- IV) Reduced cycle time.

2. Intangible benefits:

The benefits which cannot be seen or tasted or measured are called intangible benefits. Intangible benefits includes...

- I) Customer satisfaction
- II) Reduced cost of quality
- III) Improved vendor performance
- IV) Better information accuracy
- V) Fast communication
- VI) Helps in decision making.

Failure of ERP

- 1) Resistance from users itself.
- 2) Unaware about the benefits of implementing ERP.
- 3) Insufficient training to the user.
- 4) When business requirements are clearly defined.
- 5) When business processes are changed.
- 6) When users are not competent to learn ERP.
- 7) ERP solution selected is not appropriate for the organization.
- 8) Over reliance on the claims of ERP software vendors.

ERP Implementation

- For implementation of ERP waterfall mode approach is used. After getting the confirm order the project starts with meeting of organization and vendor considering organizational issues. Project is studied jointly by vendor and project in-charge.

Requirement, Definition and description (RDD)

- Firstly, the requirements of users are listed out. These requirements are data, information, function, process or reports. These requirements are studied carefully. Based on these requirements the ability of ERP solution is evaluated. Necessary changes in standard ERP packages are done to fulfill the requirements.
- On examining the fulfillment of requirements, a new documents prepared call as deviation RDD. The changes are to be checked approved by authorized person

Advantages of RDD

- I) Users of ERP get committed to the solution.
- II) Resistance to the change is eliminated.(Changes are possible)
- III) It provides revised specifications clearly.
- IV) It helps in accessing the work load due to changed requirements.
- V) Process design becomes easy.
- After establishing the RDD and testing it with the standard RDD of ERP solution, DRDD(Deviation RDD) is prepared for changes in solution and implementation.

ERP Implementation Steps

1. Meeting with user to understand ERP process.
2. Identification of RDD & DRDD.
3. Procuring (receiving) resources for the changes.
4. Training to the users.
5. Testing of ERP solution on recommended platforms.
6. Shifting of manual system to ERP solution(data transfer, opening balance).
7. Check for integrity of ERP solution by taking reports, checklists and verification.
8. Check for system performance (Speed, response).
9. Review of ERP systems performance.
10. Finalization of ERP implementation with some changes if any.

EMS and MIS:-

- In an organization management's requirement is information support from MIS rather than support in functional management. Decision making support is another requirement. Also varieties of MIS reports are needed for top level management, middle level management & front – level management. The focus area of each level is different.
- ERP solution is an integrated solution. It has capability to meet the organizational requirements since it is supported by variety of tools. ERP provides executive information and helps to formulate the organization strategies. ERP also used to locate, define and resolve problems.
- ERP is equipped with workflow automation, EDI, data warehousing & EIS which enables the workgroup management efficiently & effectively. ERP has the capability to modify MIS. The MIS design can be modified to improve decision making effectively.

ERP SELECTION

Since, the market offer a number of ERP packages, the buyer has a choice to make. Each product has its own USP and differs in a number of ways in content, scope, an ease of implementation, etc. The selection can be made on three dimensions, viz. **the vendor, the technology, the solution scope, and architecture.**

Vendor Evaluation Factors

1. Business strength of the vendor.
2. Product-share in total business of the vendor.
3. R. & D investment in the product.
4. Business philosophy of the vendor.
5. Future plans of the vendor.
6. Market reach and resource strength of the vendor.
7. Ability to execute the ERP solution.
8. Strength in the other technology knowledge and the ability to use them

9. Perspective plan of the ERP improvement with technology development.
10. Image in the business and in the information technology world.
11. Financial strength of the vendor to sustain and handle the business and technology risk.
12. Organization for product development and support.
13. The global experience of the vendor and commitment to the product for long term.

Technology Evaluation Factors

1. Client server architecture and its implementation-two tier or three tier.
2. Object orientation in development and methodology.
3. Handling of server and client based data and application logic.
4. Application and use of standards in all the phases of development and in the product.
5. Front end tools and back end data base management system tools for the data, process presentation management.
6. Interface mechanism; Data transfer, real time access, OLE/ODBC compliance.
7. Use of case tool, screen generators, report writers, screen painter and batch processor.
8. Support system technologies like bar coding, EDI, imaging, communication network.
9. Down loading to PC based packagers, MS-Office, lotus notes, etc.
10. Operating system and its level of usage in the system.
11. Hardware-software configuration management.

ERP Solution Evaluation Factor

1. ERP fit for the business of the organization in terms of the functions, features and processes, business scope versus application scope and so on.
2. The degree of deviation from the standard ERP product.
3. Ease of use; Easy to learn, implement and train.
4. The ability to migrate to the ERP environment from present status.
5. Flexible design.
6. The level of intelligent usage of help error messages, dictionaries.
7. The ability for a quick start on implementation.
8. Versatility of the solution for implementation on a platform with the project of saving the investment.
9. Rating on performance, response and integration.
10. Product quality in terms of security, reliability, and precision in results.
11. Documentation for system handling and administration.
12. Product rating in its class of products.
13. Solution architecture and technology.

The methodology of selection will begin first with the study of organization in terms of the business focus, critical application, sensitive business process, etc. Since, the ERP solution is a tool to change the style of business management; it requires thorough understanding of the business, the business issues, the management criticalities, and the socio-cultural factors. Such a study will help find out if the ERP is fit for the organization. It is a very important to find out that the ERP is fit or not, as it is the most important and critical success factor.

The price of the ERP package is difficult to judge and often it is a negotiable point in favor of the buyer in competitive scenario

The whole process of ERP implementation(in detail)

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- ❑ The price of the ERP package is difficult to judge and often it is a negotiable point in favor of the buyer in competitive scenario. Since the ERP implementation is a two three years project, the ERP solution will sustain and be adequate for the current and the future business needs for a period of five to seven years. After that, it would become a platform for the future expansions and growth.
- ❑ It is advisable for the organization to form a committee for selection of the ERP SOLUTION. IT SHOULD HAVE IMPORTANT FUNCTIONAL HEAD, A STRONG Information Technology person and a person from corporate planning function. The committee should be headed by a CEO or his designated authority. This committee should prepare a requirement document spelling out the business goals, and objectives, the futuristic scenario of business, the critical functions, processes, business focus and customer deliverables.
- ❑ A note on the management philosophy, procedures, practices and style will be a valuable input. When such a document is ready, the selected ERP vendors should be called for seeking the ERP offer. The document should be given to the vendors, and they should be allowed to study the organization and its business. All the vendors should be asked to submit a technical proposal explaining the fit of the ERP to the organization. The submission of the vendors should be scrutinized by the committee for short-listing.
- ❑ The short-listed vendors then should be asked to give the product presentation to the selected group of decision makers to seek their opinion on the product. When the product presentation is over, product demonstration should be arranged, for a detailed security and evaluation. In this process, the committee should confirm whether the critical requirement of business, in terms of information, process handling facilities, features, etc. are available or not. If some of them are not available then there is a possibility of work around to achieve the same result.
- ❑ A second evaluation note should be made for a comparative analysis of the ERP solutions and then a critical evaluation of this analysis should lead to the choice list. Simultaneously, the committee should gather information on the experience of the other organization where the ERP is implemented. This information should be on how successful the vendor is, in the implementation of the ERP? The strengths and the weaknesses of the vendor, the product and the post sales processes should be ascertained. The choice list should be weighed by these points.
- ❑ Though such an approach is appropriate, it is not always possible to bring out a clear win in the evaluation, as many factors are intangible in nature. In such an event, the committee should examine the trade off involvement in the selection. It should not happen that organizational issue dominates the choice of the ERP and in the process the best product is rejected. Ideally, the organization should be carrying out business process engineering and reengineering study, restructure the organization, modify the processes functionalities before the ERP decision is made.
- ❑ Once the committee makes the decision, the vendor should be asked to resubmit the technical and commercial proposal with price and the terms of offer. The proposal should have the following details.
 1. Scope of supply.

2. Objectives
 3. Modules and deliverables
 4. Implementation methodology.
 5. Plan and schedules of hardware and software implementation.
 6. Resource allocation
 7. Responsibility division between the organization and the vendor.
 8. Process of implementation
 9. Organization of implementation.
 10. Progress monitoring and control of the important events
 11. Process of resolving the issue all levels
 12. The official product literature
 13. Association with the other vendor its purpose
 14. Commercial submission.
 - Price by module and number of users
 - Payment terms.
 15. Process of acceptance of the ERP by stages and linking with the payments.
- ✎ Once the ERP decision is made, the vendor and organization enter into a legal contract. Such legal contract should list the obligations, duties, responsibilities, deliverables and the value components. It should also include the clauses on issues arising out of unforeseen circumstances and how to resolve them with the legal remedy available to both the parties. Since, the ERP is a product of several technologies, there should be clauses relating to safeguarding the interests of each other to cover the risk arising out of the technology failure.
 - ✎ The ERP is a tool to manage the enterprise resources to achieve the business objective. It is a supporting system and does not solve all the problems of business management. The success of the ERP lies in its implementation with commitment. It requires full participation of the organization. It is to be appreciated as a managerial tool and not as a labor saving device. Since, potentially the ERP is designed for productivity rise, the management must exploit it to its advantage by adopting the best practices or changing the practices through the business process reengineering.

ERP Solution Structure: Business Operation, Technology, Implementation

ERP Solution structure:

The Enterprise Resource Planning solution structure is built in three layers namely, technology, business and implementation

Business Operations:

On the business side, it provides solution for the data entry, data capture, transaction - processing and data base updates. It then supports an analysis of the processed result in single or the multi-dimensions for one period or successive periods. It offers a support on the comparative analysis, the budget versus the actual, the target versus the actual, the standard versus the actual and so on. The analysis also throws light on the exception conditions, for immediate attention and action.

The ERP solution provides the decision making capabilities either built-in or through the decision support systems when it comes to implementation. It provides the database functionalities for data and information management. It then has the front end tools to develop the application in an exclusive manner.

The ERP solution then handles the interfacing requirement to the legacy or the third party systems as a support to the main ERP solution. Using various information technology tools and application development tools, the application logic is developed to the business functionality. The tools are-report writers, interfacing tools, query processing tools, application development tools, the object oriented tools, etc.

Technology

The technology side of the ERP solution is managed through the database management technology for data acquisition to data base creation, updation and maintenance. The application development is done through the client/server technology, where the servers handle the specific or the general functions as the case may be and the client play the role of processing interactively and locally for meeting the information needs. The client/server implementation could be two tier or three tier, based on the design and the implementation strategy. To reduce the development effort and for ease of maintenance, the development approach is an object-oriented approach, where the class and the object libraries are built for reusability of the object and its code.

Implementation

The ERP implementation is multi-user and calls for the network usage for the work flow, communication and the access to the database (s) which may be at one location or distributed. The successful implementation of the ERP calls for a strong technology component appropriate to the environment.

Implementation involves the following topics.

- ❑ ERP's scope usually implies significant changes to staff work processes and practices
- ❑ Generally, three types of services are available to help implement such changes consulting, customization, and support.
- ❑ Implementation time depends on business size, number of modules, customization, the scope of process changes, and the readiness of the customer to take ownership or the project.
- ❑ Modular ERP systems can be implemented in stages.
- ❑ The typical project for a large enterprise consumes about 14 months and requires around 150 consultants.
- ❑ Small projects can require months; multinational and other large implementations can take years.
- ❑ Customization can substantially increase implementation times.